

PRESIDENCY UNIVERSITY

School of Artificial Intelligence and Advanced
Computing



CSS7102- Mini Project Review-2

CIVIC-VOICE An AI .Assistant for Multi-Channel Citizen Feedback Analysis in Public Services

Batch Number: CAI-20

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20231CAI0117

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Problem Statement Number: 40

Organization: Public Service Organizations (Government Agencies)

Category : Software

Problem Description:

Public agencies receive citizen feedback through emails, social media, and surveys but often analyze these in silos. An AI assistant that unifies feedback, detects themes, and highlights emerging issues would improve responsiveness. The solution should focus on sentiment and topic modeling across channels, equity-aware treatment of minority voices, and dashboards for officials to track concerns over time.

Difficulty Level: Medium–High

Abstract

:

CiviVoice AI is an AI-powered assistant designed to analyze citizen feedback received through multiple channels such as emails, surveys, social media, and online feedback platforms. Public agencies often handle these sources independently, making it difficult and time-consuming to identify recurring complaints, citizen sentiment, and emerging public-service issues. The proposed system integrates heterogeneous feedback into a unified platform and applies Natural Language Processing (NLP) techniques for text preprocessing, sentiment analysis, and topic detection. The system further identifies frequently occurring and emerging issues, assigns priority levels, and presents the results through an interactive dashboard. By converting unstructured citizen feedback into structured and actionable insights, CiviVoice AI aims to support public-service officials in understanding citizen concerns, prioritizing critical issues, and improving responsiveness and decision-making.



Github Link

Github Link: <https://github.com/inchara-k-r/AI-Assistant-for-Multi-Channel-Citizen-Feedback-Analysis>



OBJECTIVE

- **Multi-channel integration**

To develop a unified platform for collecting and integrating citizen feedback from at least three channels, such as email, surveys, and social media.

- **Sentiment analysis**

To automatically classify citizen feedback into positive, neutral, and negative sentiment categories using NLP-based machine learning/transformer models.

- **Topic and issue detection**

To identify recurring service-related topics and emerging issues from large volumes of unstructured citizen feedback using topic-analysis techniques.

- **Issue prioritization**

To assign priority levels to identified issues based on factors such as sentiment severity, frequency, and urgency, enabling officials to focus on critical concerns.

- **Decision-support dashboard**

To develop an interactive dashboard that visualizes feedback trends, sentiment distribution, major topics, emerging issues, and priority complaints for public-service decision-making.

Analysis of Problem Statement

Technology Stack Components:

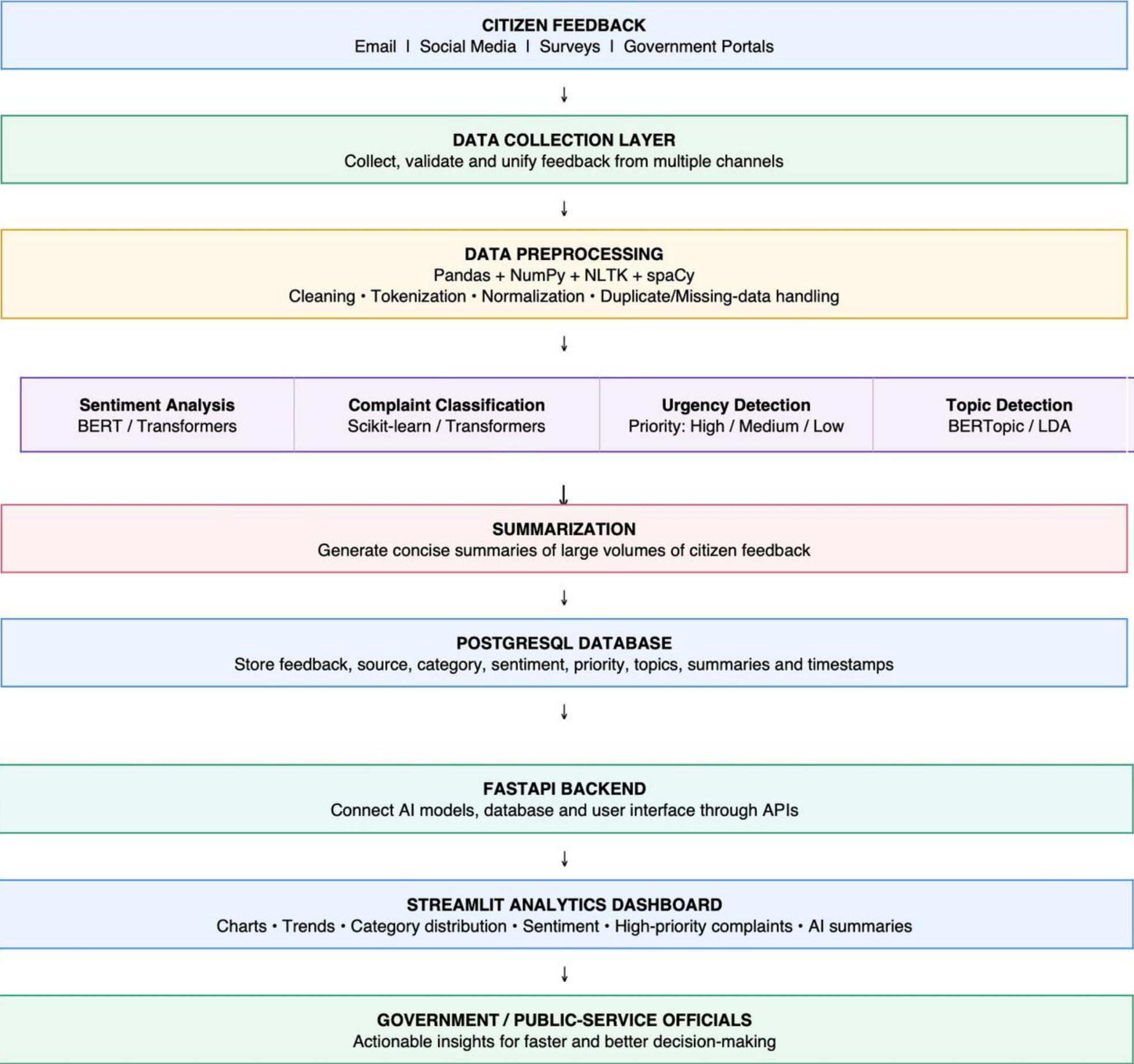
Component	Technology	Purpose
Frontend	Streamlit / HTML, CSS, JavaScript	User interface and dashboard
Backend	Python, Flask (or FastAPI)	API development and application logic
Programming Language	Python	Core development
NLP & AI	Hugging Face Transformers, BERT, NLTK, spaCy	Text preprocessing, sentiment analysis, summarization
Machine Learning	Scikit-learn	Complaint classification and model evaluation

Analysis of Problem Statement (contd...)

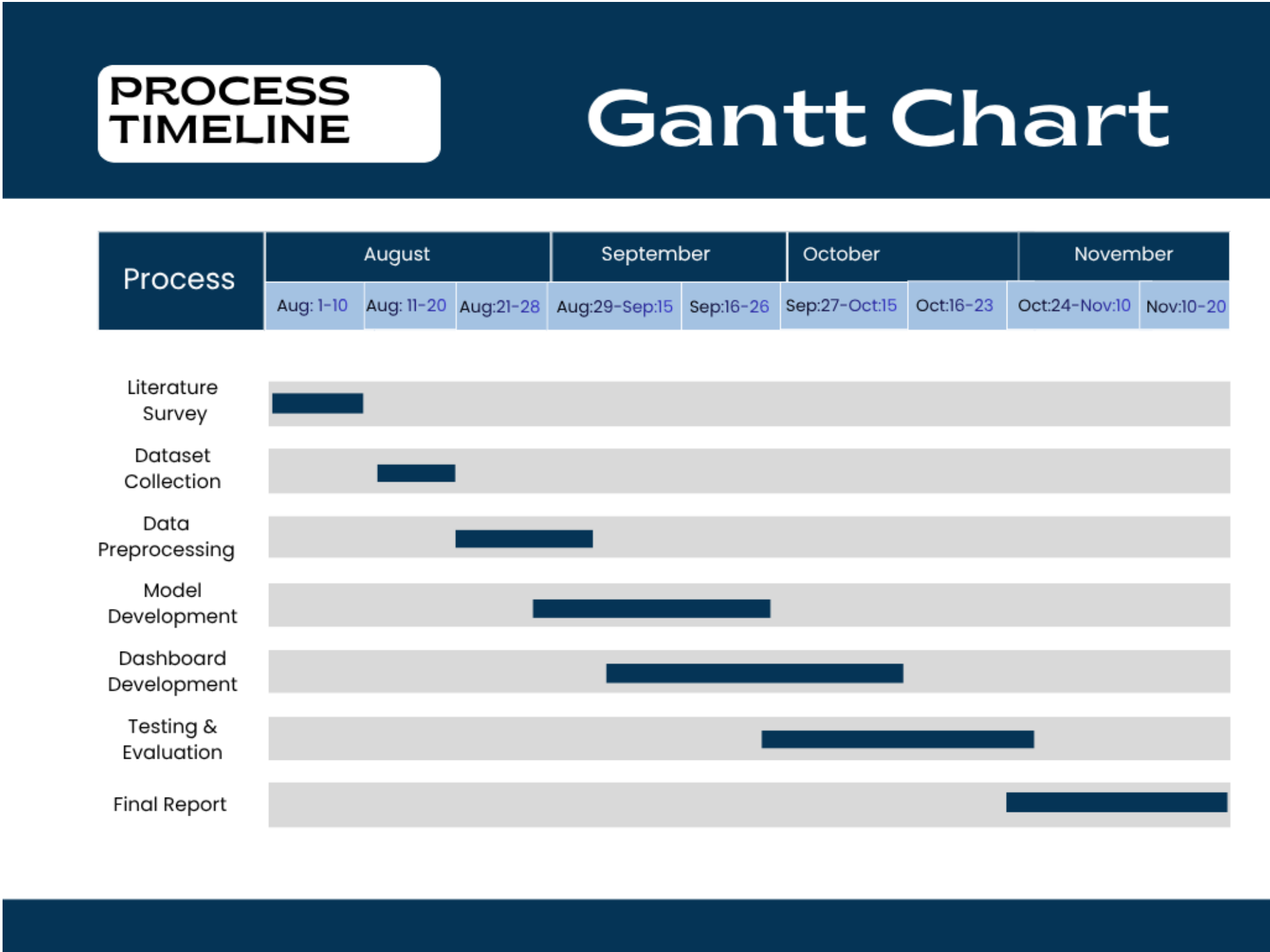
Component	Technology	Purpose
Database	MongoDB	Store citizen feedback and analysis results
Data Processing	Pandas, NumPy	Data cleaning and manipulation
Visualization	Plotly, Matplotlib	Graphs, charts, and analytics
Development Environment	Jupyter Notebook, VS Code	Model development and coding
Version Control	Git & GitHub	Source code management and collaboration
Deployment (Optional)	Render / Railway / Streamlit Cloud	Deploy the web application

Project Architecture

AI Assistant for Multi-Channel Citizen Feedback Analysis in Public Services



Timeline of the Project (Gantt Chart)



References (IEEE Paper format)

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- *G. A. Sandag, E. H. E. Soegiarto, L. Laoh, A. Gunawan and D. Sondakh, "Sentiment Analysis of Government Policy Regarding PPKM on Twitter Using LSTM," 2022 4th International Conference on Cybernetics and Intelligent System (ICORIS), Prapat, Indonesia, 2022, pp. 1-6, <https://doi.org/10.1109/ICORIS56080.2022.10031474>.*



Thank
You!

